

Section 4.3 – 4.4 Exercises

Directions: For exercises 1 - 10, rewrite each equation in exponential form.

1. $\log_4(q) = m$

2. $\log_a(b) = c$

3. $\log_{16}(y) = x$

4. $\log_x(64) = y$

5. $\log_y(x) = -11$

6. $\log_{15}(a) = b$

7. $\log_y(137) = x$

8. $\log_{13}(142) = a$

9. $\log(v) = t$

10. $\ln(w) = n$

Directions: For exercises 11 - 20, rewrite each equation in logarithmic form.

11. $4^x = y$

12. $c^d = k$

13. $m^{-7} = n$

14. $19^x = y$

15. $x^{-\frac{10}{13}} = y$

16. $n^4 = 103$

17. $\left(\frac{7}{5}\right)^m = n$

18. $y^x = \frac{39}{100}$

19. $10^a = b$

20. $e^k = h$

Directions: For exercises 21 - 30, solve for x by converting the logarithmic equation to exponential form. Leave all answers exact.

21. $\log_3(x) = 2$

22. $\log_2(x) = -3$

23. $\log_5(x) = 2$

24. $\log_3(x) = 3$

$$25. \log_2(x) = 6$$

$$26. \log_9(x) = \frac{1}{2}$$

$$27. \log_{18}(x) = 2$$

$$28. \log_6(x) = -3$$

$$29. \log(x) = 3$$

$$30. \ln(x) = 2$$

Directions: For exercises 31 - 36, use the definition of common and natural logarithms to simplify.

$$31. \log(100^8)$$

$$32. 10^{\log(32)}$$

$$33. 2\log(.0001)$$

$$34. e^{\ln(1.06)}$$

$$35. \ln(e^{-5.03})$$

$$36. e^{\ln(10.125)} + 4$$

Directions: For exercises 37 - 47, evaluate each logarithm without using a calculator.

$$37. \log_3\left(\frac{1}{27}\right)$$

$$38. \log_6(\sqrt{6})$$

$$39. \log_2\left(\frac{1}{8}\right) + 4$$

$$40. \log(10,000)$$

$$41. \log(0.001)$$

$$42. \log(1) + 7$$

$$43. 2\log(100^{-3})$$

$$44. \ln(e^{\frac{1}{3}})$$

$$45. \ln(1)$$

$$46. \ln(e^{-0.25}) - 3$$

$$47. 25\ln\left(e^{\frac{2}{5}}\right)$$

Directions: For exercises 48 - 52, evaluate each expression using a calculator. Round your answer to three decimal places.

$$48. \log(0.04)$$

$$49. \ln(15)$$

50. $\ln\left(\frac{4}{5}\right)$

51. $\log(\sqrt{2})$

52. $\ln(\sqrt{2})$

Directions: For exercises 53 - 57, state the domain and range of each function.

53. $f(x) = \log_3(x + 4)$

54. $h(x) = \ln\left(\frac{1}{2} - x\right)$

55. $g(x) = \log_5(2x + 9) - 2$

56. $h(x) = \ln(4x + 17) - 5$

57. $f(x) = \log_2(12 - 3x) - 3$

Directions: For exercises 58 - 62, state the domain, range, and vertical asymptote of each function.

58. $f(x) = \log_b(x - 5)$

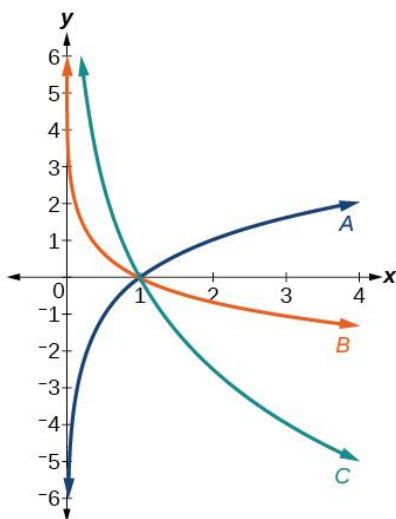
59. $g(x) = \ln(3 - x)$

60. $f(x) = \log(3x + 1)$

61. $f(x) = 3\log(-x) + 2$

62. $g(x) = -\ln(3x + 9) - 7$

Directions: For the exercises 63 - 65, match each function in the graph below with the letter corresponding to its graph.

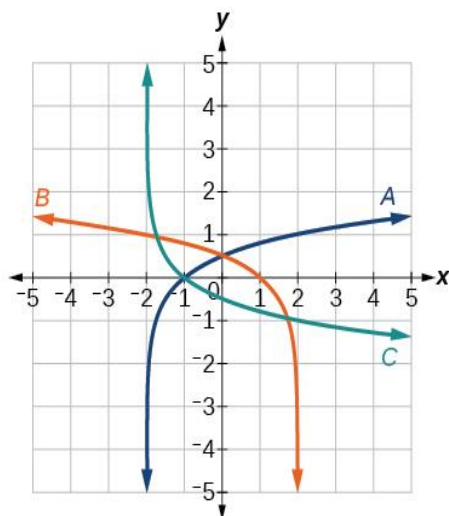


63. $f(x) = \log_{\frac{1}{3}}(x)$

64. $g(x) = \log_2(x)$

65. $h(x) = \log_{\frac{3}{4}}(x)$

Directions: For exercises 66 - 68, match each function in the graph below with the letter corresponding to its graph.



66. $f(x) = \log_4(-x + 2)$

67. $g(x) = -\log_4(x + 2)$

68. $h(x) = \log_4(x + 2)$

Directions: For exercises 69 - 70, sketch the graph of the indicated function.

69. $f(x) = \log_2(x + 2)$

70. $f(x) = \ln(-x)$